THE DIAGONAL OF THE FP FLAT PROPER SPACES STACK IS REPRESENTABLE BY ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma algebraic-lemma-representable-diagonal (canonical tag 045G). cited at proof line 3541. Criterion (2) for representability of the diagonal.
- Proposition proposition-Mor (canonical tag 0D1C). cited at proof line 3560. The Mor functors are algebraic spaces.

Source reference. Original lemma 045G: sources/stacks-chow/tags.txt, source lines 5435–5435. Cited at proof line 3541: criterion (2).

Source reference. Original proposition 0D1C: sources/stacks-chow/tags.txt, source lines 16342–16342. Cited at proof line 3560: Mor functors algebraic.

Lemma 0.2. *The diagonal*

$$\Delta : \mathcal{S}paces'_{fp,flat,proper} \longrightarrow \mathcal{S}paces'_{fp,flat,proper} \times \mathcal{S}paces'_{fp,flat,proper}$$

is representable by algebraic spaces.

Proof. We will use criterion (2) of Algebraic Stacks, Lemma 045G. Let S be a scheme and let X and Y be algebraic spaces of finite presentation over S, flat over S, and proper over S. We have to show that the functor

$$Isom_S(X,Y) : (Sch/S)_{fppf} \longrightarrow Sets, \quad T \longmapsto \{f : X_T \to Y_T \text{ isomorphism}\}$$

is an algebraic space. An elementary argument shows that $Isom_S(X,Y)$ sits in a fibre product

$$\begin{array}{ccc} Isom_S(X,Y) & \longrightarrow & S \\ \downarrow & & \downarrow{\scriptstyle (\mathrm{id},\mathrm{id})} \\ Mor_S(X,Y) \times Mor_S(Y,X) & \longrightarrow & Mor_S(X,X) \times Mor_S(Y,Y) \end{array}$$

The bottom arrow sends (φ, ψ) to $(\psi \circ \varphi, \varphi \circ \psi)$. By Proposition 0D1C the functors on the bottom row are algebraic spaces over S. Hence the result follows from the fact that the category of algebraic spaces over S has fibre products. □

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